

```
In [3]: print("Hello, Big Data!")
```

Hello, Big Data!

Part 1. Variables & Data Types

Exercise 1.1

```
In [1]: my_name="Abdul Salam"
my_age = 21
my_gpa = 14.75
am_i_present = True

print(my_name, type(my_name))
print(my_age, type(my_age))
print(my_gpa, type(my_gpa))
print(am_i_present, type(am_i_present))
```

Abdul Salam <class 'str'>
21 <class 'int'>
14.75 <class 'float'>
True <class 'bool'>

Exercise 1.2 — Type conversion

```
In [2]: age_text = "21"
age_next_year = int(age_text) + 1
print(age_next_year)
```

22

Question 1.3 — Think about it

```
In [4]: print("3" + "4")
```

34

```
In [6]: ##The + operator on two strings performs concatenation since there are strings not
```

Part 2 · Strings — Working with Text

```
In [7]: print(f"I am {my_name}, {my_age} years old, studying at AUCA.")
```

I am Abdul Salam, 21 years old, studying at AUCA.

```
In [8]: course = "Big Data Analytics"
```

```
print(course.upper())
print(len(course))
print(course[0:3])
```

Part 3 · Lists — Your First Descriptive Statistics

```
In [9]: scores = [72, 85, 91, 64, 78]

print(scores[0])
print(scores[-1])
print(len(scores))
print(sum(scores) / len(scores))
print(max(scores))
print(min(scores))
```

```
72
78
5
78.0
91
64
```

```
In [10]: scores = [72, 85, 91, 64, 78]

scores.append(88)
scores[3] = 66

print("Updated list:", scores)
print("New average:", sum(scores) / len(scores))
```

```
Updated list: [72, 85, 91, 66, 78, 88]
New average: 80.0
```

Part 4 · Dictionaries — Labeled Data

```
In [11]: me = {
    "name": "Abdul Salam",
    "age": 21,
    "program": "Software Engineering",
    "district": "Nyarugenge"
}

print(me["program"])
```

```
Software Engineering
```

```
In [12]: me["score"] = 85
me["age"] = me["age"] + 1

print(me.keys())
print(me)
```

```
dict_keys(['name', 'age', 'program', 'district', 'score'])
{'name': 'Abdul Salam', 'age': 22, 'program': 'Software Engineering', 'district': 'Nyarugenge', 'score': 85}
```

Part 5 · Class Dataset Analysis

```
In [2]: students = [
    {"id": "AUCA001", "name": "Aline Uwase", "age": 21, "gender": "F", "program": "Software Engineering", "district": "Nyarugenge", "score": 85},
    {"id": "AUCA002", "name": "Eric Niyonzima", "age": 23, "gender": "M", "program": "Software Engineering", "district": "Nyarugenge", "score": 78},
    {"id": "AUCA003", "name": "Diane Mukamana", "age": 20, "gender": "F", "program": "Software Engineering", "district": "Nyarugenge", "score": 92},
    {"id": "AUCA004", "name": "Jean Bosco Habimana", "age": 24, "gender": "M", "program": "Software Engineering", "district": "Nyarugenge", "score": 88},
    {"id": "AUCA005", "name": "Clarisse Ingabire", "age": 22, "gender": "F", "program": "Software Engineering", "district": "Nyarugenge", "score": 75},
    {"id": "AUCA006", "name": "Patrick Mugisha", "age": 21, "gender": "M", "program": "Software Engineering", "district": "Nyarugenge", "score": 82},
    {"id": "AUCA007", "name": "Sandrine Umutoni", "age": 19, "gender": "F", "program": "Software Engineering", "district": "Nyarugenge", "score": 79},
    {"id": "AUCA008", "name": "Emmanuel Nsengiyumva", "age": 25, "gender": "M", "program": "Software Engineering", "district": "Nyarugenge", "score": 86},
    {"id": "AUCA009", "name": "Grace Uwimana", "age": 20, "gender": "F", "program": "Software Engineering", "district": "Nyarugenge", "score": 90},
    {"id": "AUCA010", "name": "Yves Ndayisenga", "age": 22, "gender": "M", "program": "Software Engineering", "district": "Nyarugenge", "score": 83},
    {"id": "AUCA011", "name": "Josiane Mukeshimana", "age": 21, "gender": "F", "program": "Software Engineering", "district": "Nyarugenge", "score": 77},
    {"id": "AUCA012", "name": "Claude Iradukunda", "age": 23, "gender": "M", "program": "Software Engineering", "district": "Nyarugenge", "score": 89},
    {"id": "AUCA013", "name": "Beata Nyirahabimana", "age": 24, "gender": "F", "program": "Software Engineering", "district": "Nyarugenge", "score": 81},
    {"id": "AUCA014", "name": "Olivier Twagirimana", "age": 20, "gender": "M", "program": "Software Engineering", "district": "Nyarugenge", "score": 76},
    {"id": "AUCA015", "name": "Chantal Uwera", "age": 22, "gender": "F", "program": "Software Engineering", "district": "Nyarugenge", "score": 87},
    {"id": "AUCA016", "name": "Fabrice Byiringiro", "age": 21, "gender": "M", "program": "Software Engineering", "district": "Nyarugenge", "score": 84},
    {"id": "AUCA017", "name": "Solange Ishimwe", "age": 19, "gender": "F", "program": "Software Engineering", "district": "Nyarugenge", "score": 78},
    {"id": "AUCA018", "name": "David Hakizimana", "age": 25, "gender": "M", "program": "Software Engineering", "district": "Nyarugenge", "score": 86},
    {"id": "AUCA019", "name": "Immaculee Uwamahoro", "age": 23, "gender": "F", "program": "Software Engineering", "district": "Nyarugenge", "score": 91},
    {"id": "AUCA020", "name": "Samuel Rukundo", "age": 20, "gender": "M", "program": "Software Engineering", "district": "Nyarugenge", "score": 80}
]
```

```
In [4]: names      = [s["name"] for s in students]
all_scores = [s["score"] for s in students]
districts  = [s["district"] for s in students]
```

5.1 Describe the class

```
In [13]: ## a
print(f"Number of student :{len(students)}");
print(f"Class Average score :{round(sum(all_scores)/len(all_scores),1)}");
print(f"The highest score is {max(all_scores)} and the lowest score is {min(all_scores)}");
print(f"The range is {max(all_scores)-min(all_scores)}");
```

```
Number of student :20
Class Average score :77.2
The highest score is 94 and the lowest score is 55
The range is 39
```

5.2 — Inspect individual records

```
In [19]: print(students[0])
print(students[-1])
print(f"{students[6]['name']} from {students[6]['district']} studies {students[6]['program']}")
```

```
{'id': 'AUCA001', 'name': 'Aline Uwase', 'age': 21, 'gender': 'F', 'program': 'Information Systems', 'district': 'Gasabo', 'attendance': 92, 'score': 85}
{'id': 'AUCA020', 'name': 'Samuel Rukundo', 'age': 20, 'gender': 'M', 'program': 'Networking', 'district': 'Nyarugenge', 'attendance': 94, 'score': 92}
Sandrine Umutoni from Kicukiro studies Information Systems and scored 94.
```

5.3 — Count by category

```
In [24]: print(f'{districts.count("Gasabo")} students came from Gasabo');
        print(f'{districts.count("Kicukiro")} students came from Kicukiro');
        print(f'{districts.count("Nyarugenge")} students came from Nyarugenge');
```

```
5 students came from Gasabo
4 students came from Kicukiro
3 students came from Nyarugenge
```

5.4 — Stretch goal: who is the top student

```
In [31]: ## Top score
        top_score= max(all_scores);
        top_position= all_scores.index(top_score);
        top_name=names[top_position];
        attendance = [s["attendance"] for s in students];
        top_attendance = attendance[top_position];

        print(f"The top student is {top_name} with a score of {top_score} and attendance of {top_attendance}");
        ## Lowest score
        bottom_score= min(all_scores);
        bottom_position= all_scores.index(bottom_score);
        bottom_name=names[bottom_position];
        bottom_attendance = attendance[bottom_position];
        print(f"The bottom student is {bottom_name} with a score of {bottom_score} and attendance of {bottom_attendance}");
```

```
The top student is Sandrine Umutoni with a score of 94 and attendance of 97%.
The bottom student is David Hakizimana with a score of 55 and attendance of 72%.
```

Part 6 . Reflection

1. No this is not a Big Data, Among the 5 V's; Volume is not yet met since the data is not large (only 21 rows).
2. Today's lab taught me how to use lists, dictionaries and reading them.